

VARAD RANE

Manhattan, NY 10027 · hello@varadrane.com · linkedin.com/in/varadrane7 · www.varadrane.com · +1 201 268 8948

SUMMARY

Platform engineer with 2+ years building scalable backend services and cloud infrastructure. Expertise in Go, Python, Kubernetes, and multi-cloud platforms (AWS, Azure). Developing agentic mindset through active use of AI-assisted engineering to accelerate development and reduce operational toil.

EXPERIENCE

Fieldworker

Feb 2026 - Present

Software Developer

Jersey City, NJ

FieldWorker is a social work case management software designed for Social Service Agencies. It allows them to improve the process of delivering in-home and community health services to their patients.

- Built an LLM-based PDF extraction pipeline for Service Detail Reports using GPT-4.1, converting complex multi-column, cross-page layouts into validated structured JSON with zero traditional OCR dependency.
- Architected a local-first RAG pipeline for clinical case note processing (Ollama, ChromaDB, sentence-transformer embeddings) running fully within an 8GB VRAM budget.
- Designed a reproducible benchmarking harness tracking latency, throughput, and quality metrics across chunking, retrieval, and quantization strategies to systematically reduce information loss.

New Jersey Institute of Technology

Sep 2025 - Dec 2025

Teaching Assistant

Newark, NJ

Supported course delivery of CS 116 (DSA with C++ - Undergraduate) and CS 630 (Operating Systems - Graduate)

- Teaching C and C++ to undergraduate students, covering memory management, pointers, data structures, and systems-level programming concepts.
- Holding weekly office hours, grading assignments, and providing 1:1 feedback on code quality and debugging approaches.
- Reinforced low-level fundamentals that directly apply to performance-sensitive backend work.

Kotak Securities

Jun 2023 - Aug 2024

Senior Developer

Mumbai, India

One of India's largest stockbroking firms, part of Kotak Mahindra Bank, serving millions of retail and institutional traders across equities, F&O, and mutual funds.

- Improved API throughput by 30% by replacing session-based auth with a token architecture, reducing per-request authentication overhead across all user-facing endpoints.
- Built an Alert Management service in .NET and SQL that cut infrastructure downtime by 15%, monitoring legacy system degradation and triggering automated notifications before failures escalated.
- Refactored shared database access patterns into a reusable .NET library, eliminating two network hops and measurably reducing application latency across 12 internal services.
- Designed a queue-based aggregation layer coordinating queries across multiple SQL databases, enabling consistent high-concurrency data retrieval for trading workflows.

Kotak Securities

Jun 2022 - Jun 2023

Junior Developer

Mumbai, India

- Built full-stack features across Angular and ASP.NET, owning both frontend components and SQL-backed API endpoints for internal trading and client-facing workflows.
- Led migration of legacy applications from .NET Framework to .NET 7.0, updating dependency chains, resolving breaking changes, and aligning authentication flows with updated security standards.
- Optimized high-frequency SQL queries for production API endpoints, identifying bottlenecks through query execution plans and reducing p95 latency in critical data retrieval paths.
- Implemented bulk data ingestion pipelines for file upload workflows, enabling standardized cross-team database integration and reducing manual data handling.

Renaissance Cell, VESIT

Jan 2021 - Aug 2021

Senior Developer

Mumbai, India

Internship in collaboration with Renaissance Cell and student body to build applications for the digital and social aspect of University.

- Built a full-stack web platform and Discord bot for on-demand student certificate distribution, handling user authentication via college email, bulk uploading of digitally-signed certificates, and real-time delivery through the Discord API.
- Reduced administrative effort significantly by replacing a manual certificate issuance process with an automated, verified digital pipeline.
- Designed automated certificate issuance and distribution pipeline processing **1500+ monthly requests** with built-in error handling and verification, ensuring reliable delivery across web and Discord platforms

SKILLS

Programming Languages: C#, SQL, Python, JavaScript, TypeScript, GoLang, C, C++, Bash

Framework & Libraries: ASP.NET Core, Node.js, AngularJS, SvelteKit, TensorFlow, OpenCV, Pandas, Tailwind CSS, SASS/SCSS

Cloud & DevOps: AWS, Azure, Docker Stack, Kubernetes, Nginx, GitHub Actions

Databases: PostgreSQL, MS-SQL, MySQL, MongoDB, Firebase

Tools: Git, Postman, Figma, Puppeteer, FFmpeg, MATLAB

EDUCATION

New Jersey Institute of Technology

Newark, NJ

Master of Science in Computer and Information Science, GPA: 3.90/4.00

University of Mumbai

Mumbai, India

Bachelor of Engineering in Electronics, GPA: 3.72 (9.32/10.0)

PROJECTS

SplitSync

<https://blog.varadrane.com/posts/splitsync>

- Built a serverless PWA for itemized bill splitting with proportional tax/tip distribution and direct Splitwise OAuth sync
- Browser-only data persistence, offline support, and installable client using Next.js and TypeScript.

PrivGuard Sentinel

<https://blog.varadrane.com/posts/privguard-sentinel>

- GitHub bot that reviews pull requests using an LLM, posting inline comments from structured JSON output.
- Handles large diffs with chunking to stay within context limits, and supports configurable severity filtering to reduce noise on low-priority changes. Built in Python with GitHub Actions integration.

Customer Churn Prediction Pipeline

<https://github.com/varadrane7/customer-churn-prediction>

- Built a churn prediction pipeline achieving ROC-AUC 0.89 using XGBoost with SHAP-based feature importance analysis.
- Includes a 5-page Streamlit dashboard for batch scoring and cost-optimized intervention targeting. Designed for business-readable output, not just model accuracy.

Lazy Curl

<https://blog.varadrane.com/posts/lazycurl>

- LazyCurl is a powerful Terminal User Interface (TUI) for HTTP API interaction, written in Go. Inspired by lazygit, it brings the ergonomics of a modern TUI to curl, adding robust session management, visual request editing, and a built-in concurrent load testing engine.
- Supports Postman Collections import/export, making it a drop-in companion to existing API workflows. Built in Go using the Bubble Tea framework.

Stream Chat Organizer

<https://blog.varadrane.com/posts/stream-chat-organizer-chrome-extension>

- Chrome extension that enhances YouTube live chat readability by injecting visual elements, checkboxes, formatting, and interaction helpers into the popup chat UI. Built with vanilla JavaScript and Chrome Extensions API.

PUBLICATIONS

Automated Crack Severity Level Detection and Classification for Surface Crack Using Deep Convolutional Neural Networks